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Saveetha Institute of Medical And Technical Sciences
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Print Ease: A Smart Campus Printing System Developed Using C++ Object-Oriented Programming and Priority Queue Management

DSA – 0108

Object oriented programming using c++

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Abstract



- Automates the campus printing process through a secure and user-friendly digital platform.
- Implements Priority Queue scheduling to process print jobs efficiently and minimize waiting time.
- Integrates authentication, print request management, cost estimation, and admin monitoring into a single system.



Introduction



- Campus printing is an essential service for students and faculty to print academic documents efficiently.
- Traditional printing systems often involve manual processing, leading to delays, long queues, and inefficient job management.
- PrintEase addresses these challenges by providing a smart, priority-based printing system that automates print requests, scheduling, and administration.



Existing System



- Manual submission of print requests.
- Long waiting times during peak hours.
- No priority-based scheduling for urgent jobs.
- Limited tracking of print order status.
- Difficult for administrators to manage multiple requests.
- Higher chances of human errors and delays.



Proposed Solution



- Secure student registration and login.
- Online print request submission.
- Automatic printing cost estimation.
- Priority Queue scheduling (Faculty > Urgent > Student).
- Delivery status and print job tracking.
- Centralized admin dashboard for monitoring and report generation.



Objectives



- To develop a smart and automated campus printing management system using C++ Object-Oriented Programming.
- To optimize print job processing using a Priority Queue for faster and priority-based scheduling.
- To provide secure authentication, accurate cost estimation, and efficient administration for better print management.



Problem Statement



- Manual print request handling leads to delays and long waiting times during peak hours.
- Lack of priority-based scheduling results in inefficient processing of urgent and faculty print jobs.
- Existing systems provide limited job tracking and administrative control, reducing overall operational efficiency.





Modules 1-Student & Authentication Management



- Student Registration
- Secure Login
- User Authentication
- Profile Management
- Password Validation
- View Personal Details
- Logout Functionality
- Store Student Information



Module 2 – Print Request Management



- Upload Print Document (PDF, DOCX, etc.)
- Select Document Type
- Choose Print Mode (Black & White / Color)
- Enter Number of Copies
- Enter Number of Pages
- Automatic Cost Calculation
- Submit Print Request
- Save Print Request History



Module 3 – Priority Queue Scheduling



- Receives print requests from students.
- Assigns priority levels to each print job.
- Stores requests using a **Priority Queue**.
- Processes higher-priority jobs before lower-priority jobs.
- Displays the current print queue.
- Updates print job status (Pending → Printing → Completed).
- Automatically rearranges the queue when new requests arrive.
- Reduces waiting time through efficient scheduling.
- Ensures fair and organized print job execution.



Module 4 – Admin & Report Management



- Secure Admin Login
- View All Print Requests
- Monitor Pending Jobs
- Start Print Process
- Update Print Status
- Manage Student Records
- View Completed Jobs
- Generate Daily Reports
- Generate Monthly Reports
- View Printing Statistics
- Search Print Requests
- Delete Completed Records (Optional)



Literature Survey



S. No	Authors & Year	Name of the Journal / Conference	Model / Technique	Dataset / Environment	Features	Performance	Strengths	Limitations
1	Aljawarneh et al., (2019)	IEEE Access	Cloud-based Print Management System	University Campus	Remote print submission, job tracking, user authentication	Improved system availability and reduced manual effort	Cloud integration allows remote access and scalability	Internet dependency and security concerns
2	Kumar & Singh, (2020)	International Journal of Advanced Computer Science and Applications (IJACSA)	Queue-Based Print Job Management (FCFS)	College Environment	Print job queue management using FCFS policy	Reduced waiting time compared to manual processing	Simple implementation and easy to maintain	Does not handle priority jobs effectively
3	Rathore et al., (2021)	Journal of Information and Optimization Sciences	Priority Scheduling using Queue Algorithms	Academic Institution	Priority-based job scheduling (High > Medium > Low)	Improved job handling efficiency and reduced delay	Handles urgent jobs effectively using priority queue	Limited to print scheduling, no user management
4	Sharma & Patel, (2022)	International Journal of Computer Applications (IJCA)	Web-Based Print Management System	Educational Organization	Web interface for print request, status tracking, and reports	Better transparency and user convenience	Web-based access and real-time tracking	Depends on web server and requires internet connectivity